SOUVIK TA

Toronto, ON souvikat@gmail.com LinkedIn GitHub Medium Portfolio

EXPERIENCE

Machine Learning Engineer Intern *Multiverse Computing* • *Toronto, ON* • *Sept 2026 – Present*

- Compressing **transformer-based foundation models** to reduce memory footprint and inference compute while preserving zero-shot accuracy.
- Developed a **time-series forecasting pipeline** for electrical grid data whose predictions feed quantum-inspired network optimisation.
- Designed a **predictive maintenance framework** for an industrial facilities client that flags faults in air handling units, estimates capital costs, and guides repair-versus-replace decisions.

PhD Researcher, AI & Machine Learning *Western University* • *London, ON* • *Sept 2021 – Aug 2026*

- Engineered a **physics-informed neural ODE** framework in PyTorch with **40–55% lower prediction error** than purely data-driven models. **[EAAI]**
- Designed a physics-constrained neural network for an industrial hydrocracking unit with **23.5% lower prediction error** than standard ML baselines. **[Fuel]**
- Implemented data cleaning and validation pipelines for **6 years of real plant data** feeding a probabilistic surrogate with uncertainty quantification. **[IECR]**
- Developed **1D-CNN** models predicting 10 crude oil properties from spectral data. **[CJCE]**

PROJECTS

ChemSimGuide: Agentic Process Simulator GitHub Demo

Google GenAI Capstone 2025, Global Top 20 Finalist (6,000+ teams)

- Built a conversational AI assistant that helps chemical engineers set up simulations in the Cantera library in plain English, closing the gap between process expertise and coding skill.
- Designed a stateful **LangGraph** agent on **Gemini 2.5 Pro** that decides each turn whether to answer, ask a clarifying question, or call a tool, then routes to the right step.
- Implemented **tool calling** with a **RAG** documentation search (ChromaDB, Gemini embeddings) and a code-generation tool that returns ready-to-run Python for each setup step.

ODE Forge: Autonomous Simulation Agent GitHub Demo

LLM Hackathon for Materials Science and Chemistry

- Built an AI agent that turns a plain-language conversation into a working differential-equation model, simulation, and interactive analysis dashboard, with no coding needed from the user.
- Designed a two-phase **LangGraph** workflow on **GPT-4o**: a research phase that builds the model with the user using **RAG** over uploaded papers, and a coding phase that writes and runs the simulation.
- Ran generated code in a sandboxed **Docker** container with a self-correction loop that reads errors, fixes the script, and retries automatically; delivered through a **Streamlit** app.

Agentic Wastewater Epidemiology System GitHub

MITACS-Funded • *II Prize, Best Oral Presentation, IIChE CHEMCON 2025*

- Built a multi-agent AI system that turns a public health question into a complete, reproducible disease-modelling analysis of wastewater data, cutting weeks of manual research and coding to minutes.
- On real data from London's Greenway treatment plant, the system detected the peak of the COVID-19 Delta wave **about 5 days before** it showed up in reported clinical cases.
- Orchestrated Researcher, Coder, and Analyst agents in **LangGraph**, grounding parameters with **RAG** over peer-reviewed papers and running generated code in **Docker** with self-correction.

EDUCATION

PhD, Chemical and Biochemical Engineering Western University, London, ON 2026
Thesis: Machine Learning Frameworks for Hydroprocessing Systems: Data-Driven and Physics-Informed Approaches

B.E., Chemical Engineering BITS Pilani, India 2020

TECHNICAL SKILLS

Machine Learning	PyTorch, Keras, Tensor Networks, Physics-Informed Machine Learning, Time-Series Forecasting
LLM & Agentic AI	LangGraph, LangChain, Model Context Protocol, Tool/Function Calling, Multi-Agent Systems, Human-in-the-Loop, RAG, Structured Outputs, Prompt Engineering
LLM Engineering	Fine-Tuning (LoRA/PEFT), LLM Compression, LLM APIs (OpenAI, Gemini), Local LLMs (Ollama), Vector Databases (ChromaDB, FAISS)
MLOps & Engineering	Python, Docker, FastAPI, Streamlit, SQL/PostgreSQL, Git, CI/CD, Azure, Data Pipelines

LEADERSHIP & WRITING

- Authored 14+ technical articles on GenAI, RAG, and agentic workflows as an AI Technical Writer on Medium.
- Directed strategic initiatives and governance as a Board of Directors member for the Society of Graduate Students (6,000+ members) at Western University (2024–2025).